MIL-P-17454B
10 MARCH 1954
SUPERSEDING
MIL-P-17454A (SHIPS)
29 May 1953
MIL-P-2723
23 August 1951
MIL-P-2725
23 August 1951

MILITARY SPECIFICATION

PUMP, CENTRIFUGAL, PORTABLE, SUBMERSIBLE

This specification has been approved by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope.—This specification covers lightweight, portable, electric-motor driven centrifugal pumps capable of operating indefinitely under water. They are used primarily for the purpose of unwatering flooded ship compartments.

1.2 Classification.—Pumps shall be of the following classes, as specified (see 6.1):

Class 1—440 volts, 3 phase, 60 cycles, alternating current.

Class 2—220 volts, 3 phase, 60 cycles, alternating current.

Class 3—115 volts, 3 phase, 60 cycles, alternating current.

Class 4—230 volts, direct current.

Class 5—115 volts, direct current.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-621—Boxes, Wood, Nailed and Lock-Corner.

QQ-A-318—Aluminum Alloy 52S; Plate and Sheet.

QQ-P-416—Plating, Cadmium (Electrodeposited).

PPP-B-601—Boxes, Wood, Cleated-Plywood.

MILITARY

JAN-P-106—Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.

MIL-P-116—Preservation, Methods of.

JAN-P-132—Packaging and Packing for Overseas Shipment—Crates; Unsheathed, Wood; Nailed (for Maximum Net Load of 2,500 Pounds).

JAN-P-197—Preservation, Packaging, and Packing of Anti-Friction Bearings and Bearing Parts.

MIL-B-233—Boxes, Repair Parts.

MIL-S-853—Steel, Corrosion-Resisting: Bars and Forgings (Except for Reforging).

MIL-S-867—Steel Castings, Corrosion Resisting Austenitic.

MIL-N-894—Nickel-Copper-Alloy; Wrought.

MIL-C-915—Cable, Cord and Wire, Electrical (Shipboard Use).

MIL-D-963—Drawings, Production (for Electrical and Mechanical Equipment for Naval Shipboard Use).

MIL-C-5424—Cable; Steel (Corrosion Resisting), Flexible, Preformed (for Aeronautical Use).

MIL-I-15024—Identification Plates, Information Plates and Marking Information for Identification of Electrical, Electronic, and Mechanical Equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:11Z
Check the source to verify that this is the current version before use.

MIL-B-15071—Book, Instruction; Preparation, Contents and Approval.
MIL-P-15424—Packaging of Hand Tools; General Specification for Domestic and Overseas Preservation.
MIL-P-16298—Preservation, Packaging, Packing, and Marking of Electric Machines Having Rotating Parts (Includes Associated Repair Parts).
MIL-P-16789—Preservation, Packaging, Packing, and Marking of Pumps, Centrifugal, Close Coupled, Navy Standard.
MIL-I-16910—Interference Measurement, Radio, Methods and Limits; 14 Kilocycles to 1000 Megacycles.
MIL-M-17060—Motors, Alternating-Current Integral HP (Shipboard Use).
MIL-A-17129—Aluminum Alloy Casting; Sand, Resistant to Salt-Water.
MIL-F-17292—Format for List of Repair (Spare) Parts for Shipboard Mechanical and Electrical Equipment.
MIL-M-17413—Motor, Direct-Current, Integral HP (Shipboard Use).
MIL-N-17506—Nickel-Copper-Aluminum Alloy, Wrought.
MIL-P-17652—Preparation for Delivery of Electronic, Electrical and Electromechanical parts.

NAVY DEPARTMENT

34F3—Fittings, Hose (Water, Steam, and Oil) (Other Than Flexible Metallic).
42B5—Bearings, Ball.

STANDARDS

MILITARY

MIL-STD-129—Marking of Shipments.

DRAWINGS

BUREAU OF SHIPS

9-S-5010-L—Stuffing Tubes, Type 21 Aluminum, 22B Brass, 22S Steel, for Electrical Cables Entering Cast Enclosures.
9000-S6202-73691—Motor Starting Switch, Watertight, Rotary Type With Fusetrons, 30 Amp., 500 Volt A. C.
9000-S6202-73692—Motor Starting Switch, Watertight, Rotary Type, Resistor Start Overload and Low Voltage Protection, 60 AMP., 115 Volt D. C. and 30 Amp., 230 Volt D. C.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.—Pumps furnished under this specification shall be a product which has been tested and has passed the qualification tests specified in section 4. (See 6.2.)

3.2 General design.—All pumps shall be of an electric motor driven centrifugal type capable of operating continuously either submerged or not submerged. Each pump shall be furnished with a control switch and 75 feet of portable cable.

3.2.1 The following minimum characteristics are required:

(a) 200 gallons per minute (g. p. m.) of sea water (S. P. 1.03) at 50 feet total head.

(b) 140 g. p. m. of sea water (S. P. 1.03) at 70 feet total head.

(c) Shut-off head not less than 90 feet.

3.2.2 Each pump shall be capable of continuous performance in accordance with the characteristics specified in 3.2.1 with the pump submerged in, or under conditions equivalent to 20 feet of sea water, with no leakage of water into the motor enclosure.

3.2.3 Each pump shall be capable of continuous operation with a total suction lift of 20 feet and a total head of 70 feet, but not necessary at the delivery specified in 3.2.1.

3.2.4 Pump units shall be designed to operate satisfactorily while inclined in any position.

3.2.5 Pump units including corrugated strainers shall be designed to permit easy passage through hole 10 inches in diameter.

3.2.6 Pump suction and discharge nozzles, strainers, and foot-valves shall be suitable for use with 2½-inch hose fittings conforming to Specification 34F3.

3.2.7 Each pump shall be fitted with a corrugated or star shaped strainer formed of perforated aluminum alloy sheet. This strainer shall completely surround the pump and shall extend nearly the full length of the pump.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:11Z
Check the source to verify that this is the current version before use.

There shall be not less than 20 corrugations around the periphery of the strainer, with the ribs running lengthwise of the pump. The strainer shall have openings ¼ inch in diameter, spaced 7/16 inch on centers, and the total area of all openings shall be not less than 125 square inches. The strainer shall be attached to the pump by screwing onto the pump suction nozzle hose threads. With the strainer attached, and the pump standing upright on the suction end, the pump shall be capable of pumping water down to not more than 1¼ inch depth before losing suction.

3.2.8 Each pump shall be furnished with a foot-valve for attachment to a 2½-inch suction hose. A separate strainer, designed for attachment either directly to the pump suction or to the foot-valve, shall also be provided. This strainer shall have the same diameter as the pump casing, shall have openings of approximately 0.110 square inches spaced not more than ¾ inch on centers providing a total area of all openings of not less than 9½ square inches. The strainer and foot-valve bodies shall have minimum wall thicknesses of 3/16 inch, and each shall be provided with standard lugs for spanner wrench.

3.2.9 The pump casing shall be directly attached to the special motor frame with an intermediate compartment with stuffing boxes or seals, as approved by the bureau or agency concerned, to prevent water entering the motor. As the pumps may be idle for periods as long as 12 months, careful consideration shall be given to the type of packing or seals employed.

3.2.10 Drain plugs, inserted substantially flush with the outer pump case, shall be provided for the removal of any accumulation of water within the enclosures. Two plugs shall be provided for the motor compartment, and two for the intermediate compartment.

3.2.11 A suitable lifting eye with an opening of at least 1 inch shall be provided at the discharge end of each pump, for lowering and raising the pump.

3.2.12 The impeller and all water passages shall be given as smooth a finish on all parts as the design, construction, and materials used will permit. All projections shall be kept to a minimum, and the design of the complete unit shall be sturdy for rough handling on shipboard.

3.2.13 The pump casing shall be provided with a suitable marking cast into the casing to indicate the correct direction of rotation.

3.2.14 The pump impeller shall be keyed to the shaft, and a locknut, with positive means of securing same, provided for holding the impeller on the shaft.

3.2.15 The net weight of the alternating-current pump, including corrugated strainer but less foot-valve, cable, and controller, shall not exceed 75 pounds.

3.2.16 The net weight of the direct-current pump, including corrugated strainer but less foot-valve, cable, and controller, shall not exceed 90 pounds.

3.3 Materials.

3.3.1 Pump casings, suction heads, impellers, foot-valve bodies and foot-valve strainers, shall be of class 3 HT3 aluminum in accordance with Specification MIL–A–17129.

3.3.2 Pump discharge nozzles shall be of corrosion-resisting steel in accordance with class III of Specification MIL–S–867. At the option of the manufacturer the body of the discharge head may be of cast aluminum as specified in 3.3.1, but in this case the corrosion-resisting nozzle shall be effectively insulated from the aluminum body.

3.3.3 Pump shafts shall be nickel-copper alloy in accordance with Specification MIL–N–894, or austenitic corrosion-resisting steel in accordance with Specification MIL–S–853.

3.3.4 The aluminum sheet for the corrugated strainers shall be in accordance with Specification QQ–A–318.

3.3.5 Shaft sleeves shall be of nickel-copper-aluminum alloy in accordance with Specification MIL–N–17506. When specifically approved by the bureau or agency concerned, corrosion-resisting steel in accordance with Specification MIL–S–867 may be used for shaft sleeves.

3.3.6 Each pump shall be furnished with a combined discharge cap and carrying handle.

3.3.7 Threaded parts shall be avoided wherever possible in the pump assembly, since thread corrosion will cause difficulties in disassembly. Through bolting or plastic sleeves shall be em-

Source: https://assist.dla.mil -- Downloaded: 202610-18T10:11Z
Check the source to verify that this is the current version before use.

ployed where practicable. Bolts, studs and nuts subject to sea water shall be of austenitic corrosion-resistant steel, class 7 of Specification MIL-S-853, and shall be coated with electrodeposited cadmium conforming to type I, class A of Specification QQ-P-416.

3.3.8 Cast alumium, sheet aluminum, corrosion-resisting steel, nickel-copper alloy, nickel-copper-aluminum alloy used in any parts not specifically listed above shall conform to the specifications for such material. (See 2.1.)

3.3.9 Where not specifically required herein to conform to specifications, materials used shall be of the highest grade for the duty required, and shall be subject to reviews and approved by the bureau or agency concerned.

3.4 Treatment of materials.

3.4.1 It is of utmost importance that all galvanic circuits be broken by some reliable means, since aluminum alloys will seriously corrode unless isolated from the more noble metals wherever salt water contact is possible. Drawings shall clearly indicate all junctions of aluminum alloys with other metals, and the proposed means of isolating such metals, all of which shall be subject to approval by the bureau or agency concerned.

3.4.2 After finish machining, all aluminum parts shall be anodized to produce a hard coating by the sulfuric acid process. The anodic film on all aluminum parts shall be not less than 0.002-inch thick.

3.4.3 The anodic film shall be uniform and continuous and free from breaks, scratches, or other damage. After anodizing, the parts shall be dried and inspected as promptly as possible.

3.4.4 Anodized parts shall be handled as little as possible and with care to prevent damage and to avoid contamination from fingers or unclean gloves. Except where contamination is known to exist, cleaning shall not be attempted between anodizing and the impregnation process specified hereinafter. Parts unavoidably soiled shall be cleaned with a suitable solvent or with a detergent of a type that is completely soluble in cold, hard water.

3.4.5 As soon as possible, after anodizing, and within 24 hours, all aluminum parts shall be thoroughly vacuum impregnated with a thermosetting epichlorhydrin resin base sealer.

3.4.5.1 Prior to impregnation of the pump impeller there shall be inserted within the hub of the impeller a bushing, not less than $\frac{5}{32}$-inch thick and extending completely through the impeller hub, of compactly wound fibre glass tape.

3.4.5.2 Prior to impregnation of the pump suction head a similar bushing not less than $\frac{1}{8}$-inch thick, of compactly wound fibre glass tape shall be inserted within the wearing-ring area of the suction head. This insert, when the suction head is finally finished, will form a nonremovable wearing-ring around and below the suction end of the impeller.

3.4.5.3 Unless otherwise specifically approved by the bureau or agency concerned the following impregnation process shall be followed:

3.4.5.3.1 After the parts are positioned in the vacuum chamber, a vacuum of not less than 29 inches of mercury shall be drawn and held for 15 minutes.

3.4.5.3.2 Slowly admit the resin to a sufficient depth to cover all surfaces to be impregnated. Release vacuum and soak for 15 minutes.

3.4.5.3.3 Remove resin from vacuum chamber slowly. Remove parts from chamber and hand. Remove excess resin from the threads with brush. After dripping ceases, invert the parts and air dry for not less than 1 hour.

3.4.5.3.4 Place in a convection type oven and bake at the temperatures and for the periods as required for the particular resin used. Allow to cool to room temperature. The impregnation and baking treatment shall be fully described on the master drawings (see note 2 of 3.9.4) and shall be satisfactory to the bureau or agency concerned.

3.4.5.3.5 After the above operation submerge each piece in epichlorhydrin resin. Repeat the draining and baking instructions of 3.4.5.3.3 and 3.4.5.3.4.

3.4.5.3.6 After cooling to room temperature repeat the above dipping and baking instructions.

3.4.5.4 After impregnating and coating, the impeller hub shall be machined to fit the shaft, including the key-way. The suction head wearing-ring area shall be machined for proper clearance of the impeller. Care shall be exer-

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:11Z
Check the source to verify that this is the current version before use.

cised in the machining not to damage the resin coating on the aluminum.

3.4.5.5 All aluminum threads, joints and gaskets shall be coated with silicone molybdenum sulphide grease.

3.5 Identification.

3.5.1 Each pump shall be provided with a nickel-copper alloy identification plate or a stainless steel identification plate in accordance with Specification MIL–I–15024. The identification plate shall be set in a depressed area in the face of the pump casing.

3.5.2 In attachment of the identification plate on the pump, care shall be taken to insure that the screws will not establish a galvanic circuit with the casing.

3.5.3 Care shall be taken so that any stamping required on the identification plate after completion of the pump, such as the manufacturer's serial number and the Government inspector's official stamp, will not break the protective coating on the aluminum casing.

3.5.4 The identification plate shall contain the following information:

(a) Manufacturer's name.
(b) Manufacturer's type or model number.
(c) Manufacturer's serial number.
(d) Rated gallons per minute.
(e) Rated total head in feet.
(f) Voltage and current characteristic of motor.
(g) Rated horsepower of motor.
(h) Rated speed (r. p. m.).
(i) Contract number.
(j) Standard Navy stock number.
(k) Blank space for Government inspector's stamp.

3.6 Electrical requirements.

3.6.1 Electrical equipment shall be suitable for operation on 115, 220, or 440 volts, 3-phase, 60-cycle alternating-current; or on 115, or 230 volts direct-current as specified in contract or order.

3.6.2 Alternating-current motors shall be in accordance with Specification MIL–M–17060 for service A motors, except as otherwise specified herein.

3.6.3 Direct-current motors shall be in accordance with Specification MIL–M–17413 for service A motors, except as otherwise specified herein.

3.6.4 Motor design characteristics shall be as follows:

(a)	Ambient temperature:	
	air	40° C.
	water	30° C.
(b)	Temperature measurement	Method No. 2 only
(c)	Enclosure	Submersible
(d)	Speed	Constant
(e)	Type	A. C.: Squirrel-cage induction, design B D. C.: Shunt or compound
(f)	Duty	A. C.: Continuous when pumping water; or idling in air pumping no water D. C.: Continuous when pumping water; ¼ hour idling in air pumping no water
(g)	Bearings	Ball
(h)	Insulation	A. C.: Class H (with class B temperature limits) D. C.: Class A or B

3.6.4.1 Ball bearings shall be in accordance with Specification 42B5, except that they shall be single row, double width cartridge type prelubricated bearings.

3.6.4.1.1 Consideration will be given to bearings as specified in 3.6.4.1 except single width, if it can be shown that the extra length required by the double width bearings would adversely affect the overall weight of the pumps. Single row bearings shall be used only on specific approval of the bureau or agency concerned.

3.6.4.2 No means shall be provided for lubricating the bearings.

3.6.4.3 The motor shall be provided with a sealed-in terminal block enclosed in a watertight terminal box. Electrical leads shall be led into the terminal box through stuffing tubes, type 21, shown on Drawing 9–S–5010–L, except that the inside diameter of the tube shall be suitable for the outside diameter of the cable specified in 3.6.5.

3.6.5 *Cable.*

3.6.5.1 Thirty feet of portable type cable shall be provided from the motor to the control switch. This cable shall be type THOF–9 in

Source: https://assist.dla.mil -- Downloaded: 202640-18T10:11Z
Check the source to verify that this is the currentversion before use.

accordance with Specification MIL–C–915 except as follows:

(a) Three uninsulated stranded metal wires shall be cabled with the three insulated conductors, one wire in each of the three filler spaces.

(b) Two of the three uninsulated wires shall be 0.0625 inch diameter steel stranded wire in accordance with Specification MIL–C–5424.

(c) The third uninsulated stranded wire shall be a standard copper conductor size 2½ (26) in accordance with Specification MIL–C–915.

(d) This cable shall comply with all requirements for type THOF cable except the flexing endurance test. The completed cable shall have a breaking strength of not less than 950 pounds.

3.6.5.1.1 The three uninsulated stranded metal wires shall be securely anchored within the pump housings in such a way that the insulated conductors will be relieved of any strain when handling the pumps by means of the cable. Also the three uninsulated wires shall act as the ground conductor, and shall be grounded together within both the pump housing and the control switch housing.

3.6.5.1.2 Forty-five feet of cable shall be provided from control switches to plugs. This cable shall be type THOF–9 for direct-current pumps, and type FHOF–9 cable for alternating-current pumps, in accordance with Specification MIL–C–915. One conductor shall be grounded within the switch housing. Plugs shall not be furnished with the pumps. The cable end shall be sealed with rubber and friction tape.

3.6.5.1.3 Conductors of cable between pumps and switches shall be coded black, white, and red, with the red conductor as the field lead of the direct-current units. From switches to plugs of direct-current units the red conductor shall be the ground conductor. From switches to plugs of the alternating-current units the green conductor of the FHOF–9 cable shall be the ground conductor.

3.6.6 *Control switches.*

3.6.6.1 Each alternating-current unit shall be furnished with a control switch in accordance with Drawing 9000–S6202–73691.

3.6.6.2 Each direct-current unit shall be furnished with a control switch in accordance with Drawing 9000–S6202–73692.

3.6.6.3 When the contract or order specifies nonmagnetic equipment, control switch enclosures including stuffing tubes shall be of nonmagnetic material having a permeability of 2 or less after fabrication.

3.7 Interference.—The motors used on the pumps shall comply with the interference requirements of Specification MIL–I–16910. This applies to both radiated and conducted interference from power cords or interconnecting cables and from the equipment itself. Compliance with this requirement shall be accomplished by proper equipment design. Any proposed system used for reduction of interference shall be approved by the Bureau of Ships before incorporation in the equipment.

3.8 Repair parts and tools.

3.8.1 Each set of repair parts and tools required by the contract or order shall contain the following parts, plus any additional parts considered by the manufacturer to be essential in successful servicing of the pumps.

Parts	*Quantity*
Armature, or rotor, complete with shaft and impeller	1
Motor stator, completely wound (a. c. and d. c.)	1
Brushes (complete sets)	6
Brush rigging insulation (complete sets)	1
Brush holders (sets, complete with springs)	1
Brush holder springs	8
Shaft seals (complete sets)	2
Bearings (complete sets)	1
Bearing loading springs (complete sets)	1
Packing (complete sets)	2
Gaskets (complete sets)	2
Terminal packing rings (complete sets)	2
Connection box cover	1
Pump casing drain plugs (complete sets)	2
Can silicone molybdenum sulphide grease (¼ to ½ pound)	1
Set of control switch repair parts (see Note 2)	1
Set of tools (see Note 2)	1
Instruction books (see 3.10)	2

Note 1.—Control switch repair parts shall be furnished in accordance with drawings referenced in 3.6.6.

Note 2.—All special tools and wrenches required for maintenance of the equipment shall be furnished. No ordinary tools or wrenches shall be furnished.

3.8.2 Unless otherwise specified in the contract or order, each set of repair parts and tools shall be furnished in a repair parts box conforming to type M of Specification MIL–B–233.

Source: https://assist.dla.mil -- Downloaded: 202640-18T10:11Z
Check the source to verify that this is the currentversion before use.

3.8.3 When the contract or order specifies nonmagnetic equipment, the repair parts boxes shall be type W in lieu of type M, and all box hardware shall be nonmagnetic. In this case special tools and wrenches shall be made of nonmagnetic material.

3.9 **Master drawings.**

3.9.1 Drawings shall conform to Specification MIL-D-963 for master drawings, and shall include both outline drawings and certification data.

3.9.2 The object of the master plan system is to preclude the necessity of repeating details of individual parts, or of repeating complete drawings. A contractor shall be required to make drawings only for those classes of pumps for which he has qualified. After a master drawing has received final approval, it will be unnecessary to repeat that drawing on future contracts. Certification data sheets however are required for each class of pump on each contract or order.

3.9.3 Master drawings and outline drawings 28 inches wide are preferred.

3.9.4 Master drawings shall be furnished as follows, as applicable:

(a) One drawing (or more if necessary) of details of all parts common to all pumps. (See following notes.)

(b) One drawing detailing all parts common to all a. c. pumps.

(c) One drawing detailing all parts common to all d. c. pumps.

(d) One drawing for each class of pumps, showing details applying to the individual classes only.

(e) One drawing for each motor for each class of pumps. These drawings shall be in accordance with drawing requirements of the applicable motor specifications.

(f) One sectional assembly drawing for each class of pump. Each of these drawings shall contain a complete list of material for pump and motor, and also a complete reference table of drawings. When the same part is used in more than one class of pump, such a part shall always bear the same piece or item number in all lists of material.

(g) One outline drawing for each class of pump. This drawing shall include a dimensioned outline of pump, strainer, and control switch. It shall also contain the following:

(1) Complete pump performance data.

(2) Complete pump performance curves (curves from actual tests data.) (See 4.6.3.)

(3) Complete motor characteristics and performance data.

(4) Table of weights.

(5) Complete list of repair parts as required by Specification MIL-F-17292. (See 3.8.1.)

(6) Drawing reference table, referring to appropriate sectional assembly drawing and the standard drawing covering the control switch.

(7) Reference to letters granting qualification approval, and approval of master drawings.

(8) Standard Navy stock number of pump.

Note 1.—Details of parts such as bolts, nuts, small springs, and bearings, may be omitted provided complete identification is given. This information may be shown in tabular form on the sectional assembly drawings.

Note 2.—Complete description of the impregnation and treatment of each part shall be specified on the drawing where the part is detailed. If this involves repetition of lengthy descriptive matter, then another master drawing may be provided covering only the description of impregnation and treatment of parts. In this case the detail of each treated part shall reference this drawing citing specific paragraphs and instructions.

Note 3.—Contract numbers and vessel numbers shall not be referenced on detail master drawings, sectional assembly drawings, nor outline drawings.

Note 4.—It is not necessary for the pump manufacturer to redraw standard drawings of the control switches or any details thereof.

3.9.5 Certification data shall be furnished on size A sheets on tracing cloth. Certification data shall contain the following data plus any additional data required by the individual contract or order:

(a) Identification of equipment at head of sheet, for example:

"Certification Data
for
Portable Submersible Pumps
Class 1, 440 volts, A. C."

(b) Government (bureau or agency) drawing and revision number.

(c) Contractor's drawing and revision number.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:11Z
Check the source to verify that this is the current version before use.

(d) Specifications applicable to the equipment and date of specification.

(e) Government contract number and date, or shipbuilder's order number and date.

(f) Contractor and contractor's order number.

(g) Contractor's identification of equipment.

(h) Number of ships involved, if known.

(i) Ships identification numbers, if known.

(j) Number of pumps and sets of repair parts per ship, if known.

(k) Total number of pumps and sets of repair parts on contract or order.

(l) Reference contractor's and Government drawing numbers of applicable outline drawing and control switch drawing.

(m) Name of manufacturer of control switch, and pump manufacturer's and switch manufacturer's order numbers.

(n) Additional data such as remarks and notes, for example: Nonmagnetic control switch boxes, repair parts boxes and tools.

(o) Certifying signature of representative of contractor.

(p) File number and date of approval agency's letter of approval.

(q) Standard Navy stock number of equipment.

3.9.6 *Approval of drawings.*

3.9.6.1 Preliminary master drawings shall be submitted to the bureau or agency concerned for approval prior to submittal of equipment for qualification tests.

3.9.6.2 Upon conditional approval of the master drawings by the bureau or agency concerned, the pump may be submitted for qualification tests. Final approval of master drawings will not be granted until successful completion of qualification tests.

3.9.6.3 If the manufacturer receives a contract or order for pumps of a different voltage than that of the pumps on which he holds qualification, then the necessary additional detail master drawings and outline drawings shall be submitted for approval within 30 days after date of contract or order. Any equipment manufactured prior to approval of master drawings will be at the manufacturer's risk.

3.9.6.4 Three blueprint copies, or equal, of approved outline drawings shall be submitted with each bid or proposal.

3.9.6.5 Certification data shall be submitted to the bureau or agency concerned for approval within 30 days after date of any contract or order.

3.10 **Instruction books.**—Instruction books shall be type C in accordance with Specification MIL–B–15071.

3.11 **Alternates.**

3.11.1 The above requirements cover minimum performance characteristics and maximum size and weights. This specification is not intended to be restrictive as to design and materials, provided it is proved by actual tests that other designs and materials will result in equalling or bettering the above requirements. If an alternate design is proposed, then all departures from specification requirements should be thoroughly described giving reasons and advantages to be expected. Drawings submitted with the original requests need not be in master plan form; however, master drawings are required prior to qualification approval.

3.11.2 Every deviation from design or materials specified herein shall have specific approval of the bureau or agency concerned.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Qualification tests at a Government laboratory.**—Qualification tests shall be conducted at a Government laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.6.

4.1.1 Separate qualification will be required for alternating-current and direct-current pumps; however, pumps of only one voltage characteristic are required in each current type.

4.2 **Contract acceptance tests.**—Contract acceptance tests shall be conducted at the place of manufacture, shall be witnessed by the Government inspector, and shall consist of the following:

(a) Routine tests—required on each pump.

(b) Selective tests—required on selected pumps from a production run.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:11Z
Check the source to verify that this is the current version before use.

(c) Periodic tests—required on the first pump of each class and after a change in design which might affect performance or corrosion resistance.

4.2.1 *Routine tests.*—Each pump offered for delivery shall be subjected to the routine tests specified in table I to determine conformance with this specification. Nonconforming pumps shall be rejected.

TABLE I.—*Routine, selective, and periodic tests*

Description of test	Applicable test paragraph		
	Routine tests	Selective tests	Periodic tests
General	4.6.1		
Hydrostatic test	4.6.2		
Performance characteristic.			4.6.3
Four-hour operating test.	4.6.4		
12-hour operating test		4.6.12	
Temperature rise	4.6.5.2		
Operation in air		4.6.5.1	
Insulation resistance	4.6.6		
Dielectric strength	4.6.7		
Power input	4.6.8		
Shut-off pressure	4.6.9		
Mechanical balance	4.6.10.2	4.6.10.1	
Electrical	4.6.11	4.6.11	4.6.11
Weight		4.6.13	

4.2.1.1 *Repair parts.*—Motor repair parts shall be tested as required by the applicable motor specification. Repair rotors and pump impellers shall be tested in suitable pump frames.

4.2.2 *Selective tests.*

4.2.2.1 *Lot.*—A lot shall consist of a group of pumps of the same class produced under uniform conditions and over a short period of time which have successfully passed the routine tests.

4.2.2.2 *Sampling.*—Pumps shall be selected by the Government inspector for selective tests in accordance with table II.

4.2.2.3 *Tests.*—Sample pumps selected in accordance with 4.2.2.2 and table II shall be subjected to the selective tests in addition to the routine tests specified in table I. If any sample pump is found not to conform to this specification, the lot which it represents shall be rejected. A rejected lot may be resubmitted for selective tests provided the manufacturer, after having been informed of the reasons for rejection, has inspected or tested each pump in the lot for deficiencies noted and has removed all nonconforming units.

TABLE II.—*Sampling for selective tests*

Number of pumps in the lot	Minimum number of pumps to be tested
2 or 3	2
4 to 8	3
9 to 15	4
15 to 25	5
26 to 40	6
41 to 65	8
66 to 110	10
111 to 180	13
181 to 300	17
301 to 500	22
501 and over	(1)

[1] 1 out of each 20 produced.

4.2.3 *Periodic tests.*—The first pump of each class offered for delivery (except the pumps subjected to qualification tests) shall be subjected to the periodic tests specified in table I. Thereafter, one pump of each class shall be selected for periodic tests during each calendar year during which that class of pump is being offered for delivery. If the Government inspector is satisfied that the pumps conform to this specification and that materials and treatments are identical to those of the pumps which passed the qualification tests, he may waive annual periodic tests. If the Government inspector has reason to suspect that the aluminum treatment, materials or technique being used is not in accordance with: (1) this specification; (2) treatment, materials or techniques approved by the bureau or agency concerned; (3) treatment, materials, or techniques used in the pumps passing the qualification tests, then a corrosion resistance test may be required. Such tests shall be identical to the life test specified in 4.6.14. A periodic test shall be required after any change in design which may affect the performance characteristics or corrosion resistance.

4.3 **Contractor's tests.**—*Prior to the tests* specified in 4.2, the pump manufacturer shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:11Z
Check the source to verify that this is the current version before use.

conduct sufficient tests to insure that the design of the equipment conforms in all respects to this specification.

4.4 Retests.—To prevent delays, not more than two tests shall be made; the second tests shall be made within such time after the first as stipulated by the Government inspector. Failure to make the necessary repairs or to remedy defects within the stipulated time shall be considered sufficient cause for final rejection of the equipment.

4.5 Reports.

4.5.1 *Qualification tests.*—Qualification tests reports (12 copies) shall be forwarded by the laboratory to the Bureau of Ships. These reports should include not only the findings but also any comments and recommendations looking toward a more satisfactory damage control pump.

4.5.2 *Routine and selective test data.*—Routine test data and selective test data recorded at time of tests of pumps, other than periodic-tested pumps, shall be retained by the Government inspector and shall not be distributed to other activities except when specifically requested by the bureau or agency concerned.

4.5.3 *Periodic test data.*—A test record of each periodic test, including the data for routine and selective tests on the same pump, shall be recorded on standard Navy forms furnished by the Government inspector. These forms shall be either carbon-backed, inked, or otherwise prepared so that they may serve as master copies from which blueprint reproductions may be made. The master copy shall be forwarded to the bureau or agency concerned and one copy shall be furnished the Government inspector responsible for conducting the tests. No copies of the test record shall be made or distributed to any other Government activity, manufacturer, or shipbuilder.

4.6 Methods of test.

4.6.1 *General.*—The pump shall be subjected to a thorough examination to ascertain that the material, workmanship, and design are in conformance with this specification. Integrity of coating of parts shall be scrutinized. The fit of parts shall be observed with particular reference to the interchangeability of such parts as are liable to require replacement during normal service life of the pump. This inspection and examination applies to parts during every stage of manufacture.

4.6.1.1 Sampling and test of anodic films on aluminum alloys shall conform to the following:

4.6.1.1.1 All aluminum alloy parts shall be visually inspected for compliance with 3.4.3.

4.6.1.1.2 The thickness of coating may be determined by the use of separate specimens. A minimum of 2 specimens shall be used to represent each lot of the same class and anodized together. The specimens shall be between 0.025- and 0.064-inch thick and not less than 3 inches long and shall be anodized concurrently with the lot of items they represent. The material of which each test specimen is made shall be any one of the alloy designations for the particular class within which the production work is found:

Class 1—2S, 3S, 4S, 42S, 53S, 61S, 63S and all clad alloys.

Class 2—24S, 75S, all castings and other alloys not listed as class 1.

4.6.1.1.3 Instead of using separate specimens as specified in 4.6.1.1.2, two production parts may be selected at random from each lot for determination of the thickness of anodized coatings.

4.6.1.1.4 The thickness of the anodized coating shall be determined in at least four locations on each part or specimen tested. The complete lot shall be rejected if any thickness measurement does not conform to 3.4.2.

4.6.1.1.5 Either a microscopic method, the micrometer measurement method, or other suitable method may be used for thickness measurement. The micrometer measurement method consists of determining the difference in thickness of an article, in the same area, before and after anodic treatment. The micrometer measurement method shall not be used until a correlation has been developed, using the microscopic method, between the change in thickness of the specimen and the thickness of the anodic coating, as a result of the anodic treatment.

4.6.2 *Hydrostatic test.*—After machining but prior to impregnation, each pump casing shall be subjected to a hydrostatic test of not less than 50 pounds per square inch (p. s. i.).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:11Z
Check the source to verify that this is the current version before use.

A slight weeping, or sweating, will be acceptable since the later impregnation should overcome such defects; however, more than slight weeping, or weeping over a large area, may indicate a structurally unsound casting and shall be considered cause for rejection.

4.6.3 *Performance characteristics test.*—Sufficient performance data shall be taken and recorded to prepare characteristic curves from shut-off to free delivery. The following test curves shall be supplied:

(a) Capacity (g. p. m.) versus total head in feet.

(b) Capacity (g. p. m.) versus overall efficiency (motor and pump).

(c) Capacity (g. p. m.) versus brake horsepower.

(d) Capacity (g. p. m.) versus power input.

The above data and curves shall be supplied for conditions of flooded suction and 20 foot suction lift.

4.6.4 *Four-hour test.*—The pump shall be run submerged for not less than 4 hours with a total head of 50 feet. The pump delivery shall be not less than 200 g. p. m. During this test the positions of the pumps shall vary; some pumps being vertical, some horizontal and some included in a midway position. At the conclusion of this test there shall be no evidence of water in the motor compartment, and no more than a slight amount (less than a tablespoon) of water in the intermediate compartment.

Note.—Class H insulation sometimes gives off a slight amount of water vapor. Therefore a slight amount of moisture may be found in the motor compartment of class H insulated motors which is not caused by leakage. This should be not more than dampness, not water which will run out.

4.6.5 *Temperature measurement.*—The maximum temperature rise of motor windings, by method 2, shall be determined under conditions of pumping with submerged suction at 50-foot total head, and also while pumping with submerged suction and free delivery. The temperature rise over the temperature of the water being pumped shall not exceed the temperature rise permitted by the applicable motor specifications. These temperature measurements may be made at the time that the performance characteristics tests are run. (See 4.6.3.)

4.6.5.1 Immediately following the tests specified in 4.6.5, the pump shall be operated in air, with suction open to air and discharge closed. Direct-current pumps shall operate for 15 minutes without exceeding temperature limitations of Specification MIL-M-17413. Alternating-current pumps shall operate continuously, reaching a stable temperature not exceeding temperature limitations of Specification MIL-M-17060.

4.6.5.2 The temperature rise of motor windings, by method 2, shall be determined during and at the end of the four hour operating test (see 4.6.4) of each pump. This temperature rise shall not materially differ from that specified in 4.6.5, and shall in no case exceed the temperature rise permitted by the applicable motor specifications.

4.6.6 *Insulation resistance.*—Immediately following the temperature rise test (see 4.6.5.2) the insulation resistance of each unit shall be measured in accordance with the applicable motor specifications.

4.6.7 *Dielectric tests.*—Immediately following the insulation resistance tests, dielectric tests shall be run on the unit in accordance with the applicable motor specifications.

4.6.8 *Power input.*—Power input shall be measured during the four-hour test (see 4.6.4) and the twelve-hour tests (see 4.6.12) and shall not exceed by more than 1 percent the power input for the same pumping condition specified in 4.6.3. Power input shall also be measured during the test specified in 4.6.5.1.

4.6.9 *Shut-off pressure test.*—Following the four hour tests (see 4.6.4) the shut-off pressure of the pump shall be determined, and shall be not less than 90 feet.

4.6.10 *Mechanical balance.*

4.6.10.1 Mechanical balance of selected pumps, including completely assembled rotating elements, shall be measured by the instrument methods required by the applicable motor specifications.

4.6.10.2 Mechanical balance of the pump shall be checked by "touch" during the four hour tests (see 4.6.4) and during the operation in air. (See 4.6.5.1.)

4.6.11 *Electrical tests.*—All electrical, routine, selective and periodic tests shall be made

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:11Z
Check the source to verify that this is the current version before use.

as required by the applicable motor specifications except as modified herein, and except as rendered impracticable by the construction of the pumps.

4.6.12 *Twelve hour run.*—Selected pumps shall be run continuously for 12 hours under conditions equivalent to submergence in 20 feet of water and operating at a total head of 50 feet. The pumps shall be lying horizontal during these tests. At the conclusion of this test there shall be no evidence of water in the motor compartment and not more than 3 tablespoons of water in the intermediate compartment. (See note of 4.6.4.)

4.6.13 *Weight.*—Selected pumps shall be weighed to determine that the equipment is in substantial agreement with the weights shown on the approved outline drawings. Weights shall not exceed the weights shown on the approved outline drawings by more than 5 percent and in no case shall they exceed the weights specified in 3.2.15 and 3.2.16.

4.6.14 *Life test.*—The life test shall be conducted as follows after pumps have passed all the routine, selective, and periodic tests:

(a) Run continuously for 7 days submerged in sea water pumping through a 50-foot length of 2½-inch fire hose. The corrugated strainer shall be attached during this run.

(b) Remove pump from water. Do not rinse. Drain intermediate and motor compartments, noting carefully the amount of water removed, if any. Replace drain plugs. Do not disassemble pump. Allow pump to lie on the dock in the weather for 7 days.

(c) Repeat (a) and (b) alternately until approximately 1,500 hours have elapsed, not less than 800 hours of which shall be pumping.

Note.—It is not intended that the 7 day periods be exact, and they may be varied as required by routine operation of the laboratory. The total time and pumping time should, however, be as required above.

(d) Repeat routine tests.

(e) After the above operations the pump shall be completely disassembled, care being exercised during disassembly to note any evidences of corrosion (inside or outside) erosion, wear and leakage.

(f) Reassemble and return to manufacturer.

4.7 Interference tests.—Tests for determining the presence of interference shall be conducted in accordance with Specification MIL-I-16910. Tests shall be made both with the pumps submerged and unsubmerged. The manufacturer shall furnish either directly or by subcontract all facilities for performing the interference tests including test equipment, a suitable location for the performance of the tests and qualified technical personnel.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Equipment.*

5.1.1.1 *Domestic shipment, immediate use.*—The cleaning, preservation, and packaging of the equipment shall be in accordance with the manufacturer's commercial practice.

5.1.1.2 *Domestic shipment and storage or overseas shipment.*—The preservation and packaging of the equipment shall be in accordance with Specification MIL-P-16298.

5.1.2 *Maintenance parts and tools.*

5.1.2.1 *Mechanical.*—Mechanical repair parts (onboard and stock) and tools shall be preserved and packaged in accordance with Specifications MIL-P-16789 and MIL-P-15424, as applicable. Anti-friction bearings shall be preserved and packaged in accordance with Specification JAN-P-197.

5.1.2.2 *Electrical.*—Electrical repair parts shall be preserved and packaged in accordance with Specifications MIL-P-17652 and MIL-P-16298, as applicable.

5.1.2.3 *Other parts.*—The preservation and packaging of parts not specifically covered herein shall be in accordance with Specification MIL-P-116.

5.1.2.4 *Unit packaging.*—Unless otherwise specified in the contract or order, repair parts and tools shall be packaged one item per unit package unless used in sets or in quantities greater than one.

5.1.2.5 *Intermediate packaging.*—When specified in the contract or order, onboard repair parts shall be snugly packaged in repair parts boxes conforming to type M or W, as specified (see 6.1) in accordance with Specification MIL-B-233.

Source: https://assist.dla.mil -- Downloaded: 202610-18T10:11Z
Check the source to verify that this is the current version before use.

5.1.2.6 *Index list of repair parts.*—An index list conforming to Specification MIL-F-17292 shall be inserted in the repair parts box.

5.2 **Packing.**

5.2.1 *Equipment.*—The equipment shall be packed for domestic shipment immediate use, domestic shipment and storage, or for overseas shipment in accordance with Specification MIL-P-16298. (See 6.1.)

5.2.2 *Repair parts and tools.*—Unless otherwise specified in the contract or order, repair parts and tools shall be packed in cleated plywood or nailed wood boxes conforming to Specification PPP-B-601 (domestic type) or NN-B-621 for domestic shipment, and in accordance with Specification PPP-B-601 (overseas type) or JAN-P-106 for overseas shipment. The gross weight of nailed wood and wood cleated boxes shall not exceed 150 pounds. Boxes shall be strapped in accordance with Specification PPP-B-601 or JAN-P-106, as applicable.

5.2.3 *Repair parts boxes.*—When repair parts boxes are specified in the contract or order, they shall be overpacked in shipping containers conforming to Specification JAN-P-132 or NN-B-621. The gross weight shall not exceed 200 pounds.

5.3 **Technical publications.**

5.3.1 Instruction books, pamphlets, and handbooks, as applicable, shall be packaged in accordance with method IC-3 of Specification MIL-P-116. Two copies of the instruction book shall be packed in each repair parts box.

5.4 **Marking.**—In addition to any special markings required by the contract or order, interior packages as well as exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be "Pump, Centrifugal, Portable, Submersible".

6. NOTES

6.1 **Ordering data.**—Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Quantity of pumps, sets of repair parts, and number of instruction books required.

(c) Class required. (See 1.2.)

(d) If nonmagnetic equipment is required, it should be definitely stated. (See 3.6.6.3 and 3.8.3.)

(e) If repair parts boxes are required, whether type M or W are required. (See 5.1.2.5.)

(f) Whether domestic or overseas shipment is required; if domestic, the type. (See 5.1 and 5.2.)

(g) The following information should be submitted with each bid:

(1) An approved outline drawing (see 3.9.4 (g)) of each class of pump covered in the bid.

(2) A statement that the equipment proposed to be furnished will be in strict accordance with the approved outline drawings.

6.2 In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Military Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Bureau of Ships, Navy Department, Washington 25, D. C., and arrange to have the products that they propose to offer to the Army, the Navy, or the Air Force, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Navy Department, Washington 25, D. C.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Corps of Engineers
Navy—Bureau of Ships
Other interest:
Army—T.

☆U.S. GOVERNMENT PRINTING OFFICE: 1974—713-155/6684

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:11Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:11Z
Check the source to verify that this is the current version before use.